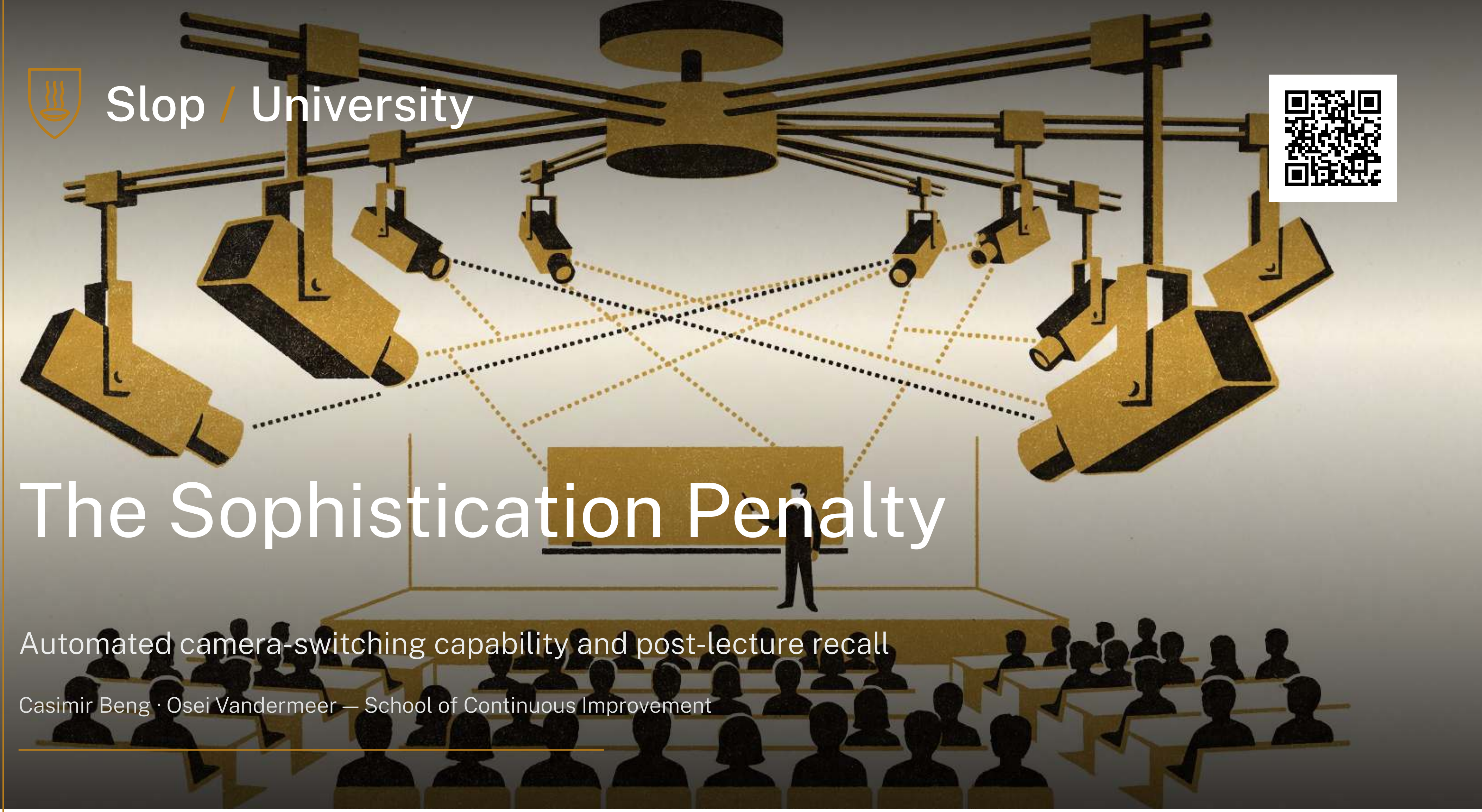

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The Sophistication Penalty

Automated camera-switching capability and post-lecture recall

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The problem

Lecture-capture rigs have grown from a static tripod camera to multi-camera, gaze-tracking, auto-framing systems. Does a rig capable of finding the whiteboard, the lecturer, and a raised hand in real time produce a recording anyone can learn from afterwards?

Research questions

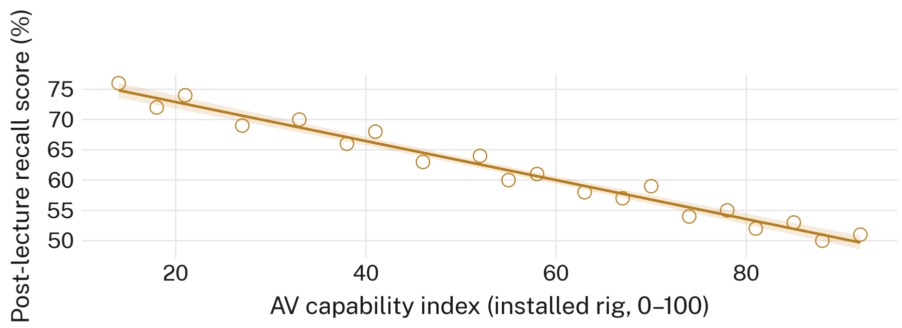
- Does installed AV-capability tier predict post-lecture recall score?
- Does automated switch frequency track self-reported disorientation?
- Is there a capability tier past which recall degrades faster than coverage improves?

Approach

- n = 214 students · 6 lecture theatres · 3 semesters (2024–2026)
- AV capability index assigned per installed rig, static single-camera to gaze-tracking auto-frame
- Recall measured by a 10-item quiz at 48h delay; switch timestamps logged directly from the AV controller, no lecturer input

Findings

Recall score falls as installed capability rises across the sampled theatres. Switch frequency, not camera count alone, carries the association most consistently.



Mean recall fell from 76% (capability index 14, static single-camera theatres, 2024 intake) to 51% (index 92, gaze-tracking auto-frame, 2026 intake) as installed capability rose ($r = -0.72$), tracking the rig-upgrade schedule exactly.

Implications

Switch frequency appears to fragment attention across the visual field rather than resolve it. A rig optimised for coverage may be optimised against comprehension. Recall loss is steepest above index 70, where auto-framing begins to anticipate the lecturer's next position rather than follow it, cutting to the whiteboard a beat before the point being made lands. Disorientation self-reports rise in step with the same threshold, suggesting the two measures track a single underlying effect rather than two independent ones. Next: a randomised-tier trial holding switch frequency constant across capability levels, to separate coverage from cutting rate.

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Recall quizzes de-identified at collection; no video retained beyond the semester of capture; thresholds pre-registered. Supported by a School of Continuous Improvement seed grant; we thank the technical staff of the monitored theatres.